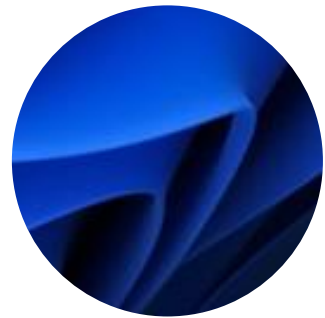


# Advanced Cisco Networking Academy - Booting Windows 11 and Installing Essential Software

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CCNA



## Purpose

This lab acts as the foundation for future lab work as an operational computer is essential for network configuration and management. By setting up the Windows 11 operating system and installing essential software, the computer will be equipped for performing tasks in a Cisco networking environment.

## Background

In the 1970's, while bonding over a shared interest in computers and programming, Bill Gates and Paul Allen got their hands on a brand-new microcomputer called the Altair 8800. Just one month after the microcomputer hit the front cover of Popular Electronics, the two had developed their own version of the BASIC programming language for it, creating the first computer language program for a personal computer. In February of 1975, they sold their program to MITS of Albuquerque, New Mexico: the first customer of Microsoft.

Microsoft gained its name by the combination of “microcomputer” and “software,” reflecting the origins of the company. Ever since its founding, Microsoft has been known for its operating systems. Specifically, in 1981, Microsoft released their famous MS-DOS. This disk operating system was the dominant OS for personal computers. It featured a command-line interface for user interaction and a hierarchical file system that included support for subdirectories. However, command-line interfaces require knowledge of complex syntax, so a more accessible system was overdue. Windows began as small updates to the MS-DOS, including a GUI extension in 1985. Later versions also included various programs such as Native Windows File Manager, Program Manager, and Print Manager. But it would be another decade before Windows kickstarted its fame.

In 1995, Windows 95 was released, fully joining Windows and MS-DOS into a standalone operating system. One of the major selling points was the system's support for the Internet. Later, in 2001, Windows XP took over. With this release, Microsoft overwrote the core software with a more powerful code base for better management and interfaces. They also developed multiple versions targeted at various customers such as consumers, businesses, and multimedia developers.

Unfortunately for Microsoft, one of their following versions of Windows gained a poor reputation for being slow, big, and taxing on people's computers. Windows Vista, made public in 2006, also released in multiple versions to accommodate various purposes. In a quick effort to fix the critiques of Vista, Microsoft released Windows 7 in 2009. This OS held onto many of the interfaces of Vista but vastly improved the issues. This new system was praised for its speed improvements, minimal system requirements, and intuitive design.

Another 3 years later, Windows 8 was released, and much like its predecessor, Vista, it was disliked by many. Most of the complaints were due to confusing UI and

Microsoft's radical changes to prioritize touchscreen devices. Despite this, Windows 8 had made many positive changes from previous generations. For instance, Windows 8 implemented the first start menu of any Windows OS. Additionally, settings could be synchronized across machines.

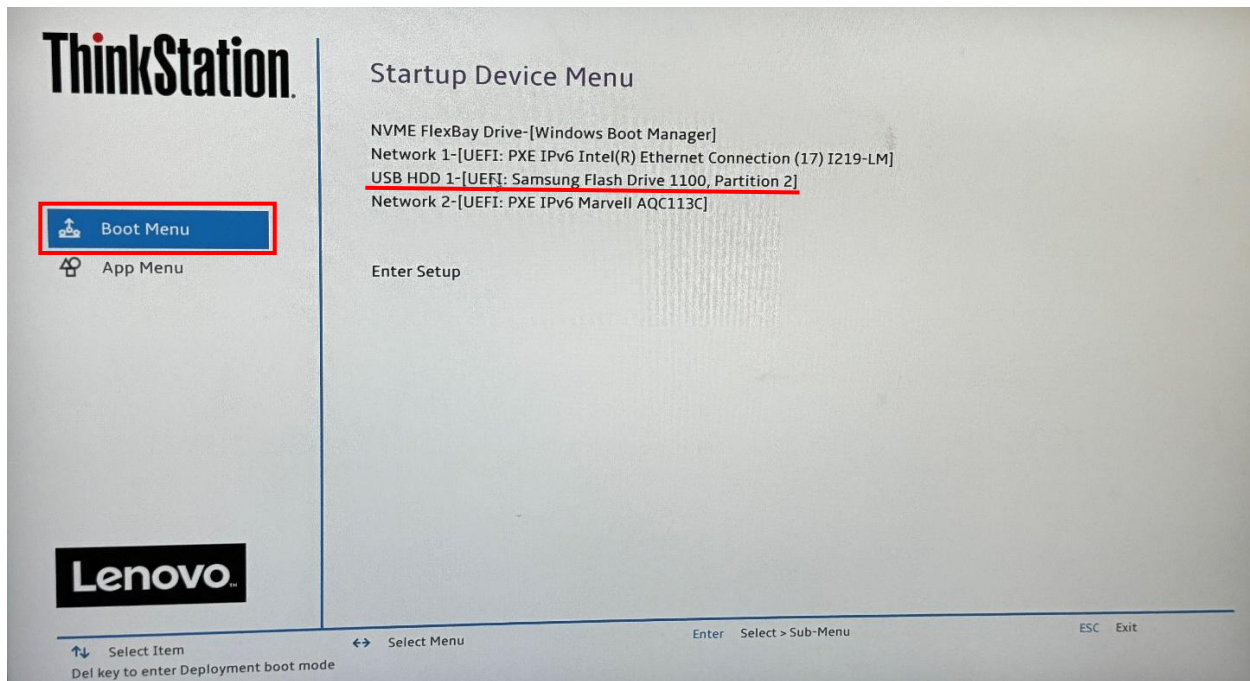
In 2014, Microsoft announced what was intended to be their final operating system: Windows 10. This version of Windows would improve upon the flaws of previous generations and aimed to solidify itself as an OS that could stand firm in an age where technology was improving at an exponential rate. To do this, features such as Cortana, a digital personal assistant akin to that of Apple's Siri or Amazon's Alexa, an improved internet browser called Microsoft Edge, and many redesigned interfaces would be included with the OS. Upon release in 2015, Windows 10 was praised for these features, and many believed it had exceeded expectations.

Yet despite statements declaring Windows 10 as the final version, Microsoft announced the brand-new Windows 11 in 2021 before releasing it in October of the same year. Unable to win against the developments in AI assistants, Microsoft created Copilot, an AI replacement for Cortana. Additionally, Microsoft redesigned many interfaces including the start menu to switch towards a more minimalistic and clean aesthetic. This, however, stirred up resistance from Windows users. Many criticized the start menu and the new taskbar layout, the somewhat restrictive hardware requirements, and the lack of intuitive and user-friendly design. But the OS has received many updates in the years following its release, and Microsoft has prioritized the fine-tuning of Windows 11 over the development of a successor.

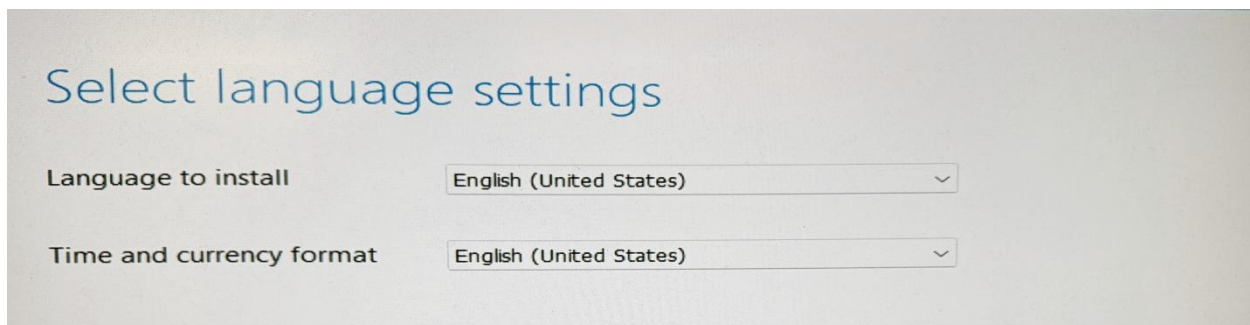
To install Windows, a bootable USB containing an ISO (a disk image of the operating system) is required. Tools like Rufus, first released in 2011, can format a flash drive and make it bootable. To start the installation, the user must boot from the USB by entering the Boot Menu by pressing a specific key during startup depending on the motherboard. Typically, this key is F12.

## **Installation**

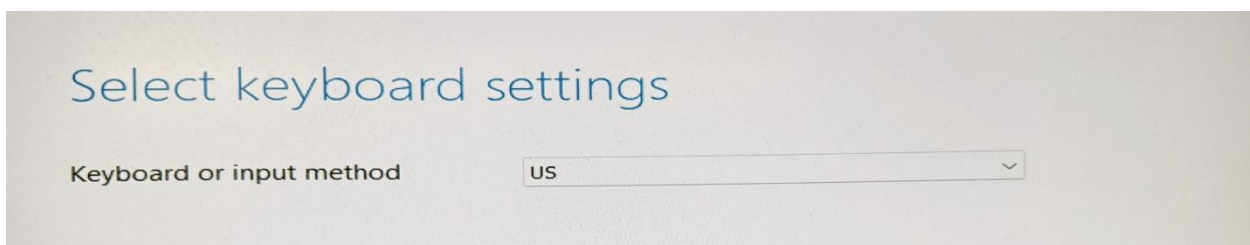
To begin a clean install of Windows 11, plug in the bootable USB into the powered-down computer. Then, start up the computer while repeatedly pressing the BIOS enter key.



Once in the BIOS, select the bootable USB inside of the Boot Menu, and setup will begin automatically.

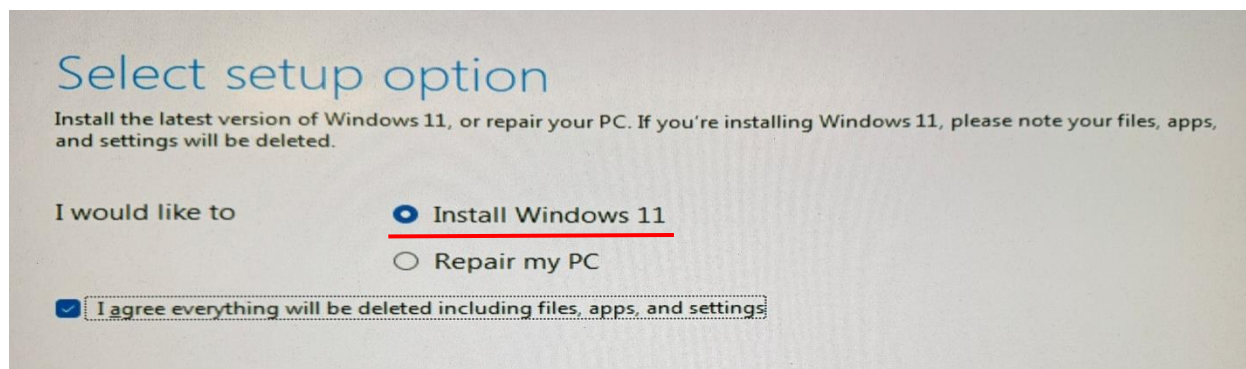


Select the desired language, time, and currency settings before pressing Next.

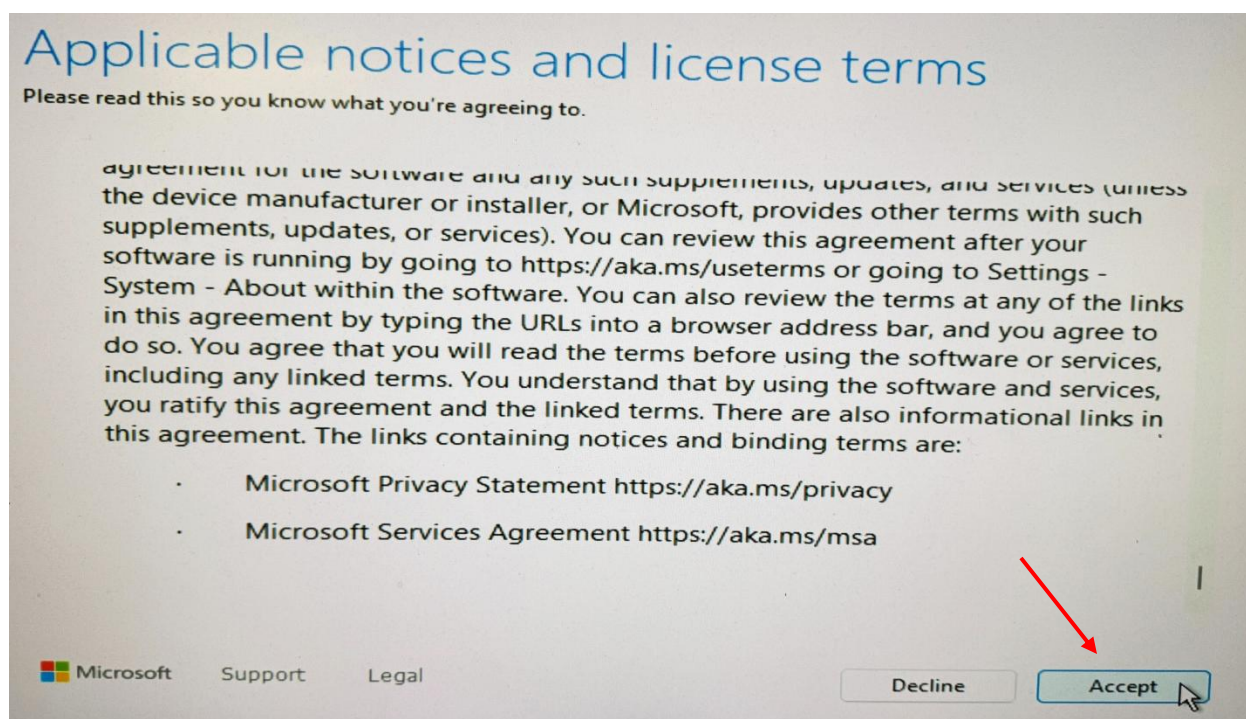


To finalize basic settings, choose a keyboard or input method layout and click Next.

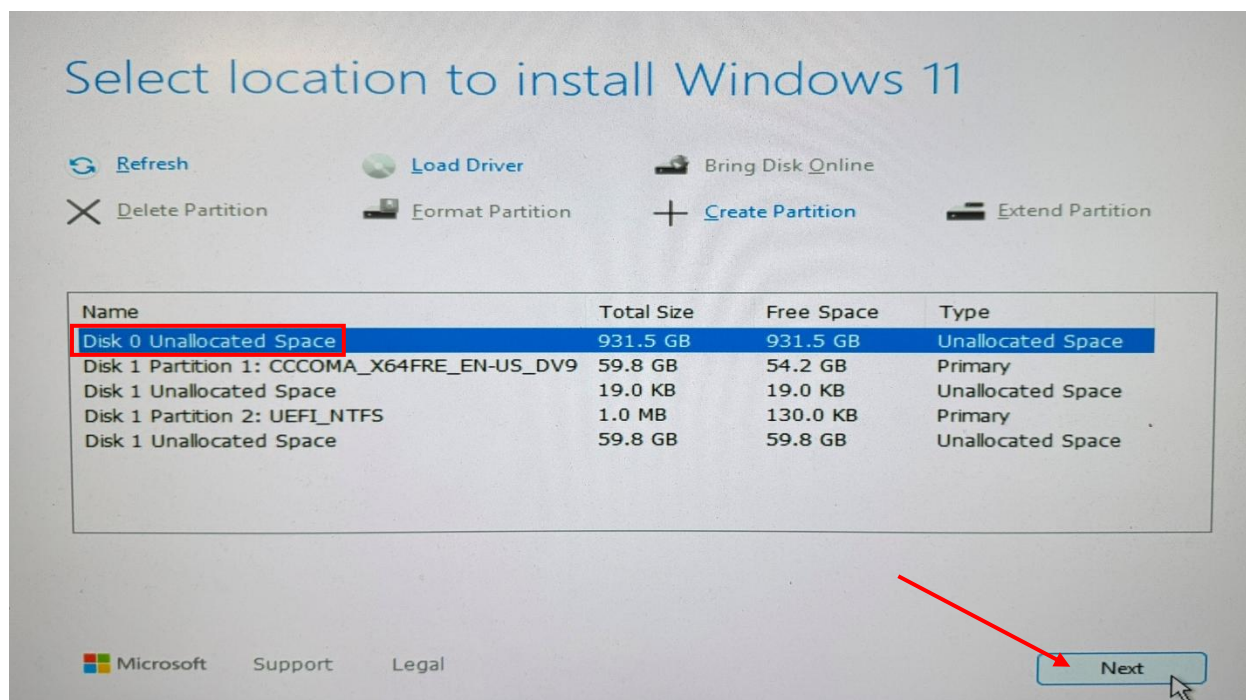




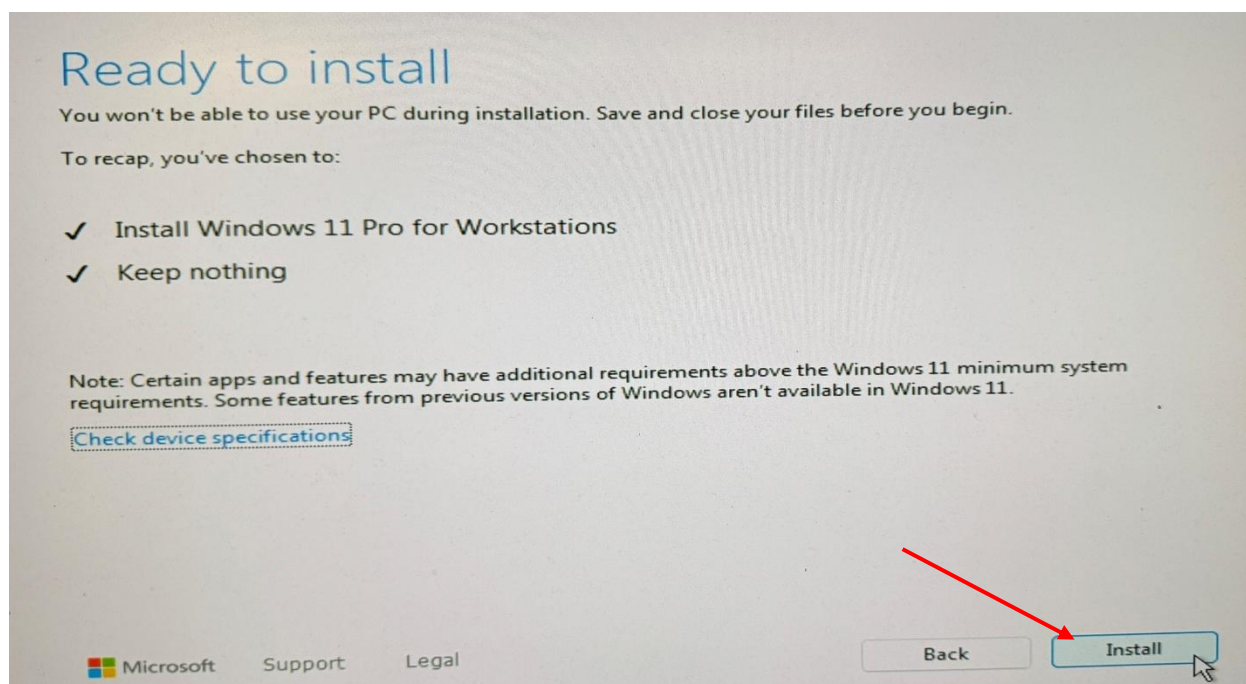
Select “Install Windows 11” to start a fresh boot of the OS. Then, click Next.



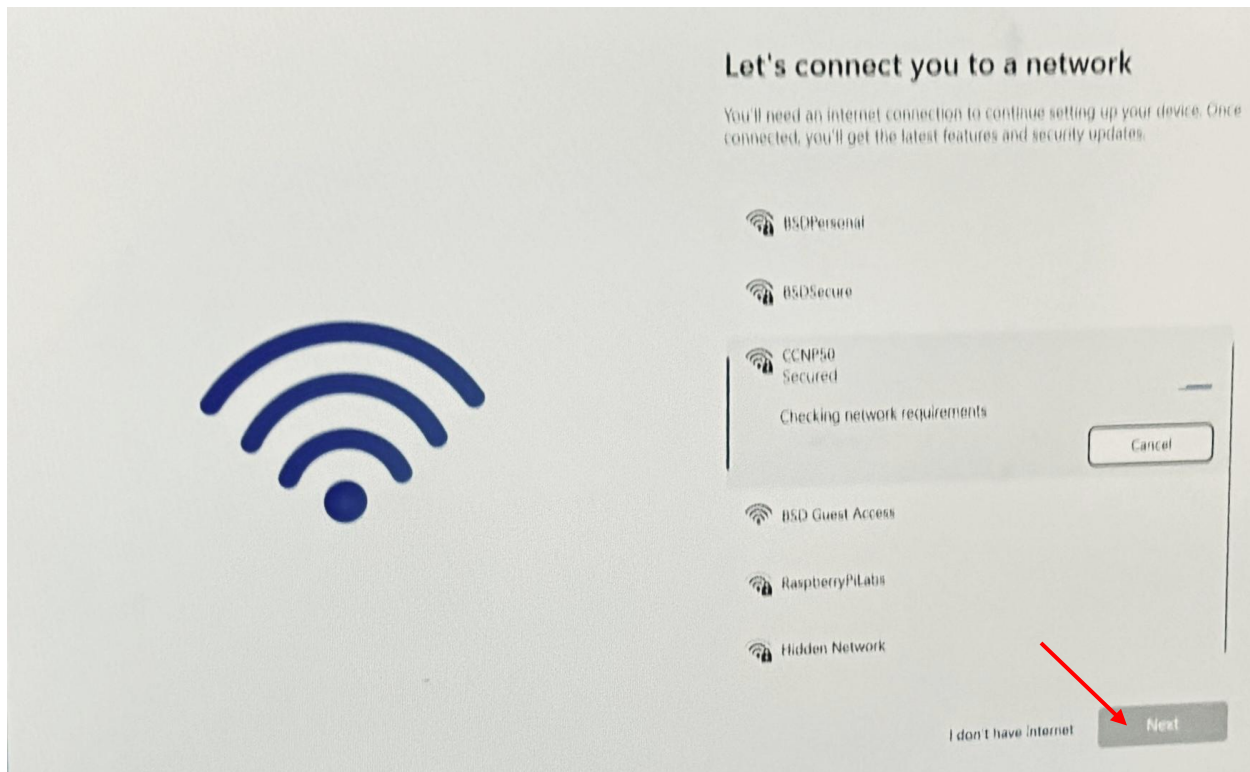
Once the page loads and you understand the agreement displayed, click Accept.



After loading, locate all the items in the list labeled Disk 0. Do not touch the other disks. If any partitions exist, select them and click Delete Partition. Once only one item named “Disk 0 Unallocated Space” exists, select it and click Next. This will enable a fresh installation of Windows 11.



Ensure the installation settings are to install Windows 11 and to keep nothing. Then, click Install.

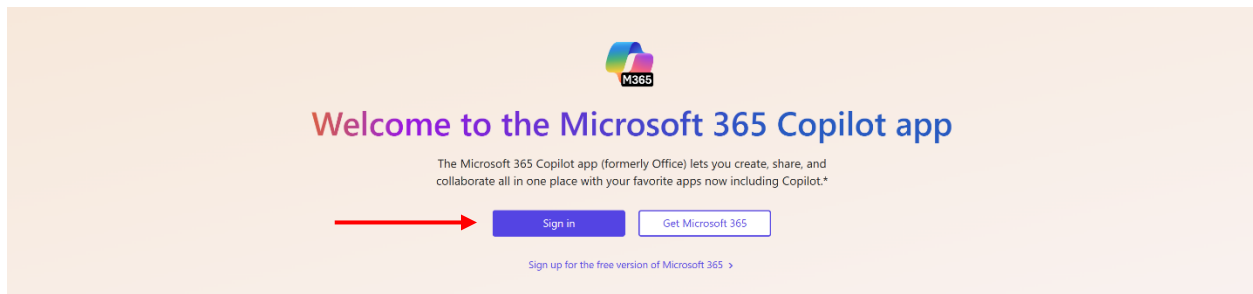


Select your Wi-Fi network and enter the password. Once connected, click Next. Follow any remaining steps to set up an account and you will be brought to a fresh Windows 11 desktop.

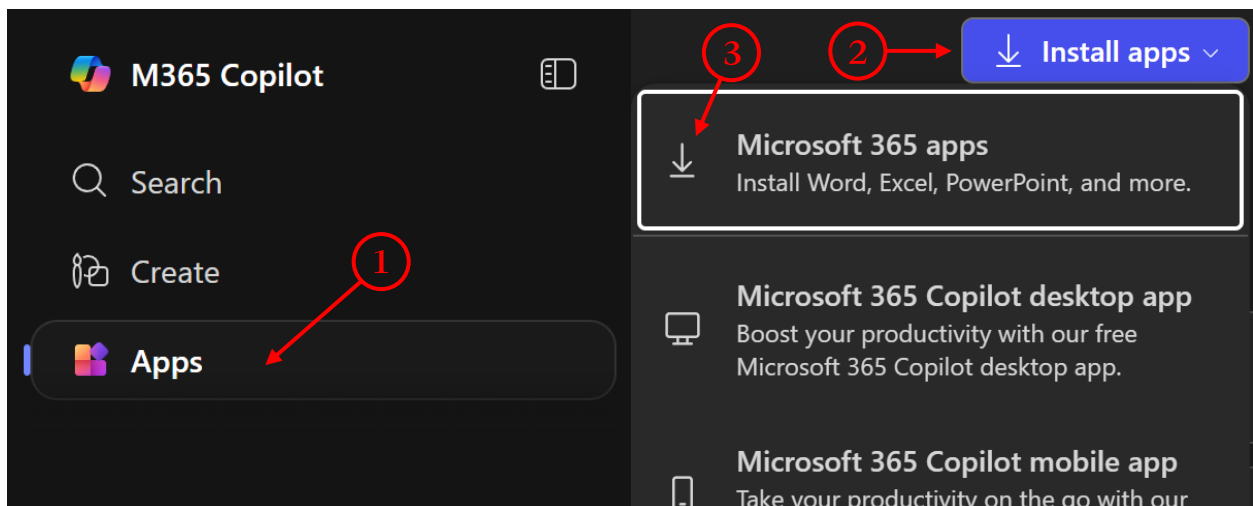


First, open the start menu or the search bar in the taskbar and find Settings. In the Settings app, navigate to the Windows Update tab at the bottom of the left-hand list. Then, click Check for Updates or Update. When the updates finish, restart your computer by clicking Restart Now.

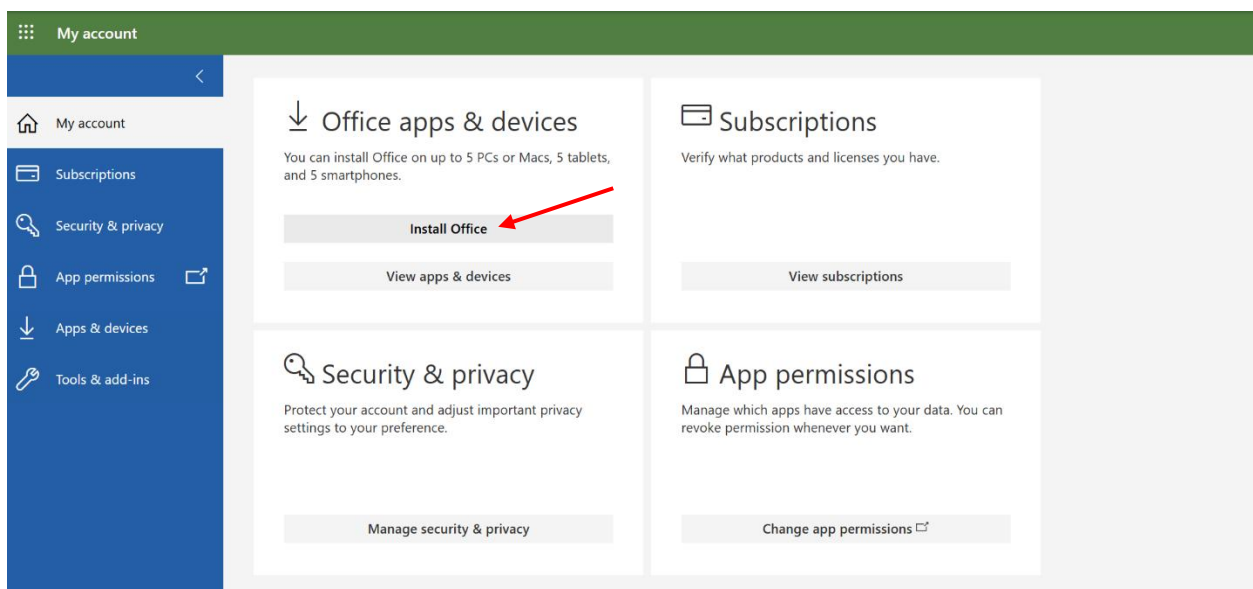




To install Microsoft 365 applications, open a web browser and navigate to office.com. Log into your Microsoft account.

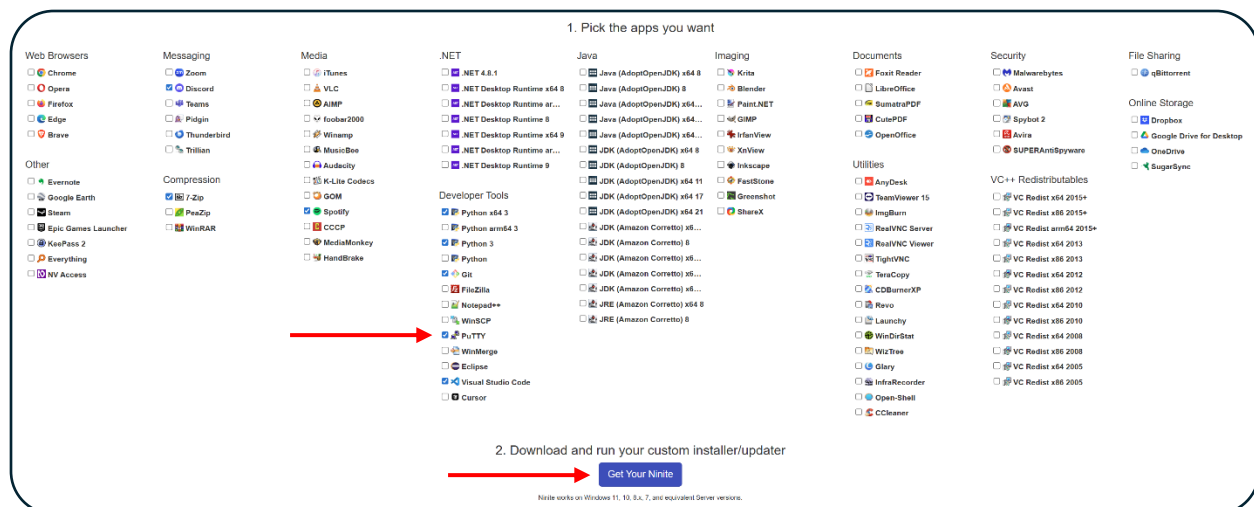


Once logged in, go to Apps, click Install apps, and select Microsoft 365 apps.

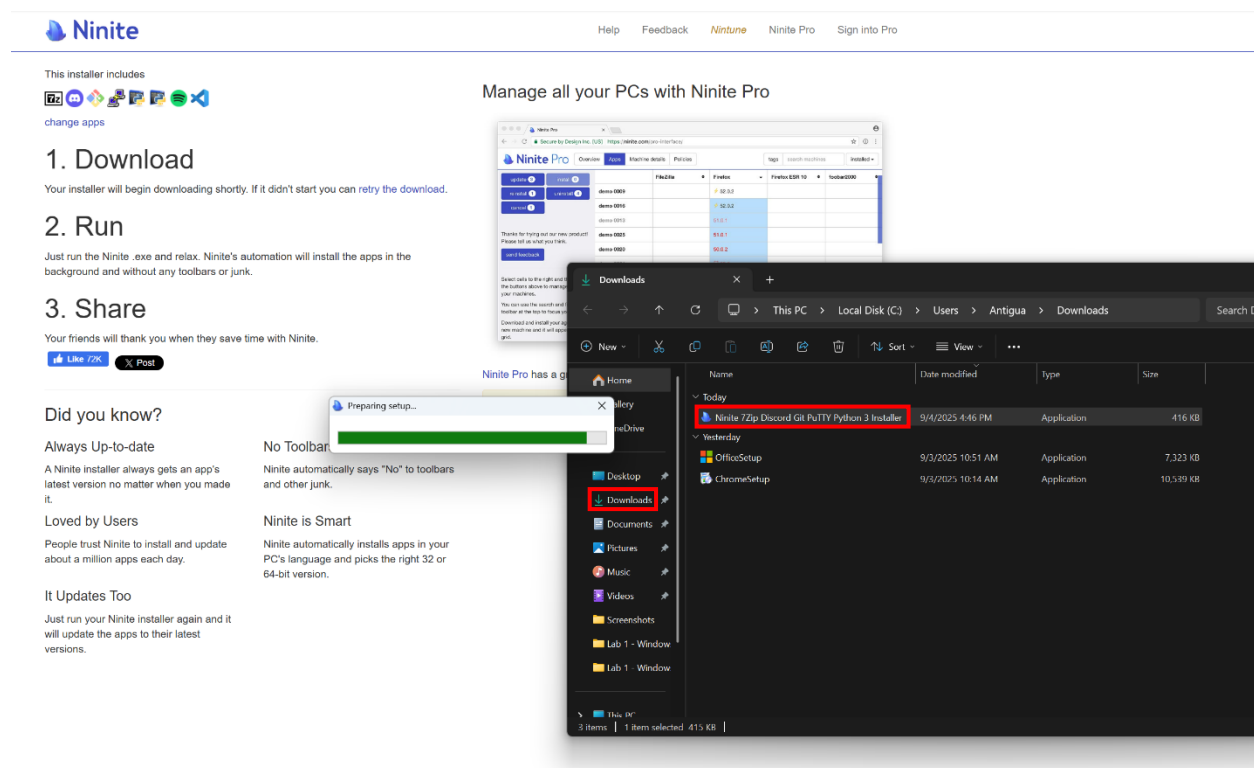


You will be brought to the download page. Click Install Office to begin installing the core Microsoft 365 apps such as Word, Excel, PowerPoint, Teams, etc.





To install many essential applications through one install, go to Ninite.com and select all of the apps you would like to install. Once ready, click Get You Ninite. For a Cisco networking environment, the necessary program from this list is PuTTY, an open-source terminal emulator and serial console for Windows.



When the download finishes, double-click the file in Downloads from File Explorer to begin installing all of the apps.

# NVIDIA App

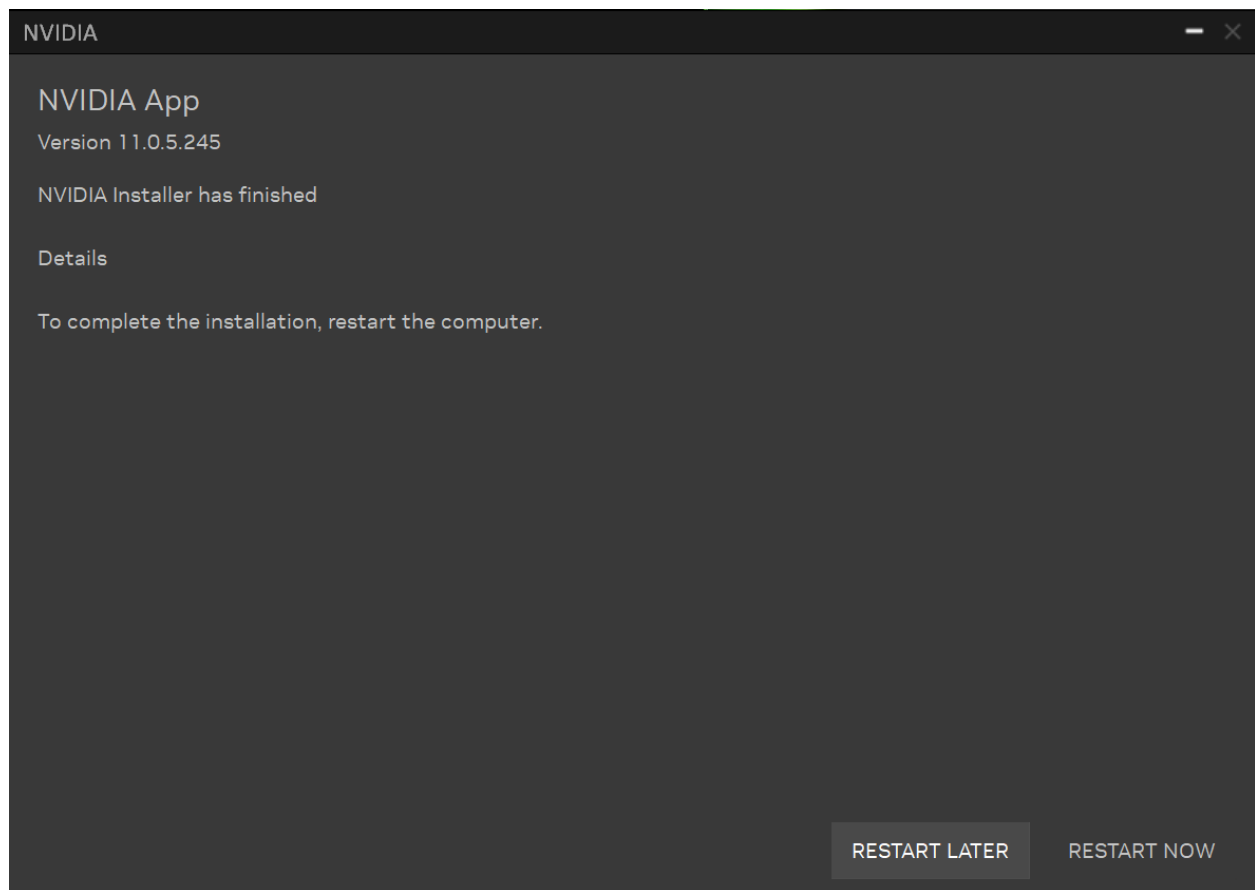
The NVIDIA App is the essential companion for PC gamers and creators. Keep your PC up to date with the latest NVIDIA drivers and technology. Optimize games and applications with a new unified GPU control center, capture your favorite moments with powerful recording tools through the in-game overlay, and discover the latest NVIDIA tools and software.



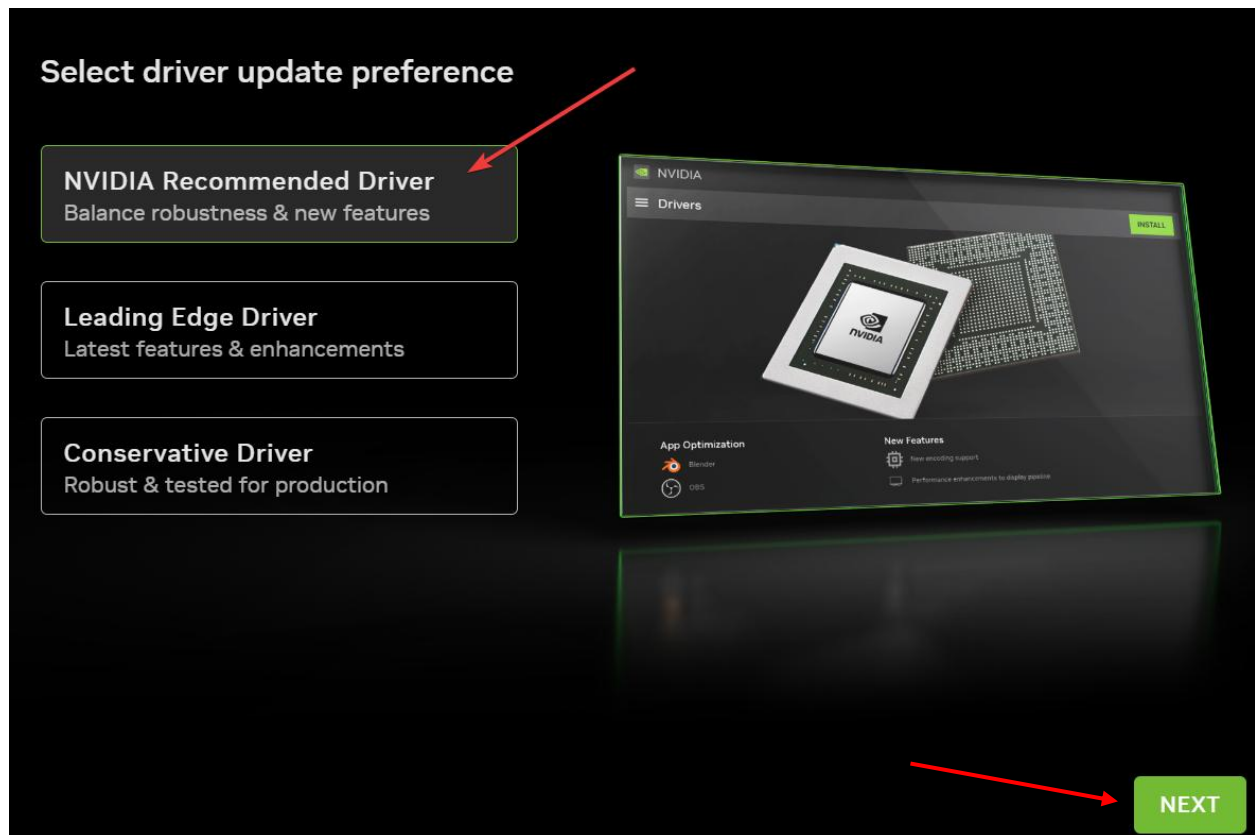
**Download Now**

**Learn More >**

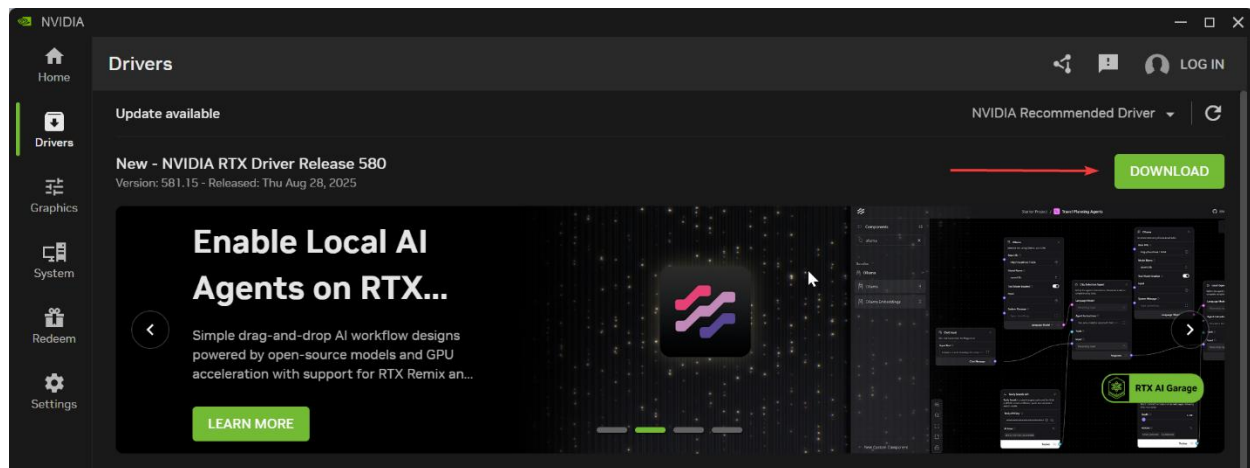
To download the latest drivers for an NVIDIA graphics card, use a web browser to go to search NVIDIA App and select the first download link to bring you to <https://www.NVIDIA.com/en-us/software/NVIDIA-app>. Click download and open the file.



After agreeing to the terms, restart the computer if requested by the app.



Open NVIDIA App and select NVIDIA Recommended Driver. Then, continue through the pop-ups.



Once able, locate Drivers on the left side and click Download.



To install the drivers, click Install.



# Getting Started with Cisco Packet Tracer

This course is part of the **Learning Collections - Cisco Packet Tracer**

Your on-ramp to Cisco Packet Tracer. Get familiar with the simulation environment and download the latest version.

 **Self-Paced Online**  
Learn online at your own pace

 **Instructor-Led**  
Learn with an academy

English (English) ▼

**Resume Course** 3,854,450 already enrolled

AVAILABLE LANGUAGES  
العربية, 中文(简体), English, Español, Français, Português, Український

For a Cisco networking environment, Packet Tracer is essential. To download, you must join the introductory course on NetAcad. Navigate to <https://www.netacad.com/courses/getting-started-cisco-packet-tracer?courseLang=en-US> and click Get Started With Self-Paced. You will be forced to login or create a NetAcad account. After, click Resume Course.

Module 1: Download and Use Cisco Packet Tracer

1.0. Install Cisco Packet Tracer 2/5

1.0.1 Video - Welcome to Getting Started with Cisco Packet Tracer

1.0.2 First Time in the Course

**1.0.3 Download Cisco Packet Tracer**

1.0.4 Why Should I Take this Course?

1.0.5 Video - Overview of Cisco Packet Tracer

1.1. The Cisco Packet Tracer Interface 6/8

Module 2: Create a Cisco Packet Tracer Network

Getting Started with Cisco Packet Tracer Course Final Exam

## 1.0.3 Download Cisco Packet Tracer



To obtain and install your copy of Cisco Packet Tracer, please follow the instructions from the link below:  
<https://www.netacad.com/resources/lab-downloads>

After logging in, navigate to section 1.0.3 and download Packet Tracer using the link.

## Learning Resources



**Cisco Packet Tracer**  
Cisco Packet Tracer, an innovative network configuration simulation tool, helps you hone your networking configuration skills from your desktop. Use Packet Tracer to experiment while building, managing & securing infrastructures.

To obtain and install your copy of Cisco Packet Tracer, please follow these simple steps:

**Step 1.**Download the version of Packet Tracer you require.

[Packet Tracer 8.2.2 MacOS 64bit](#)  
[Packet Tracer 8.2.2 Ubuntu 64bit](#)  
[Packet Tracer 8.2.2 Windows 64bit](#)

**Step 2.**Launch the Packet Tracer install program.

**Step 3.**Launch Cisco Packet Tracer by selecting the appropriate icon.

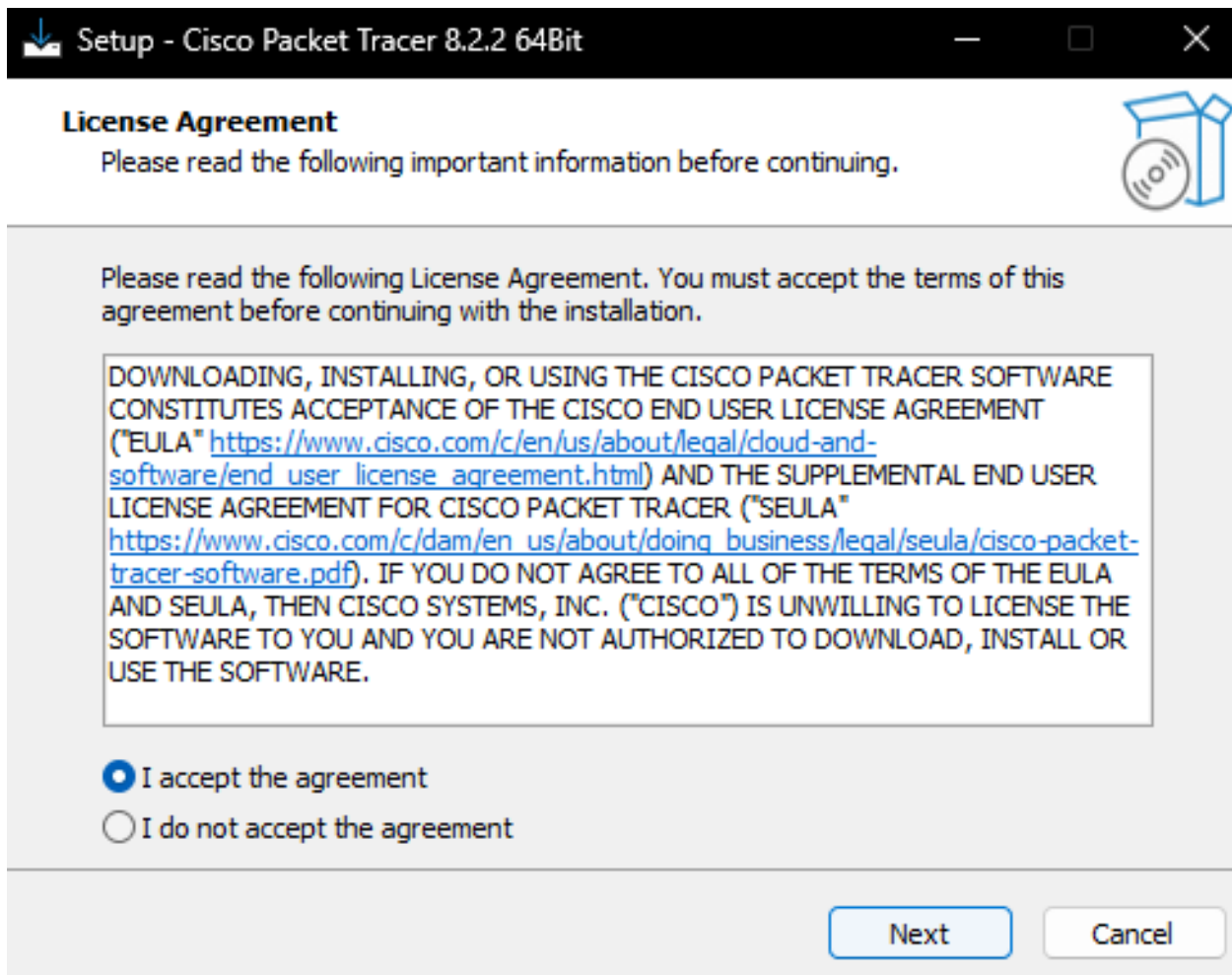
**Step 4.**When prompted, click on Skills For All green button to authenticate.

**Step 5.**Cisco Packet Tracer will launch and you are ready to explore its features.

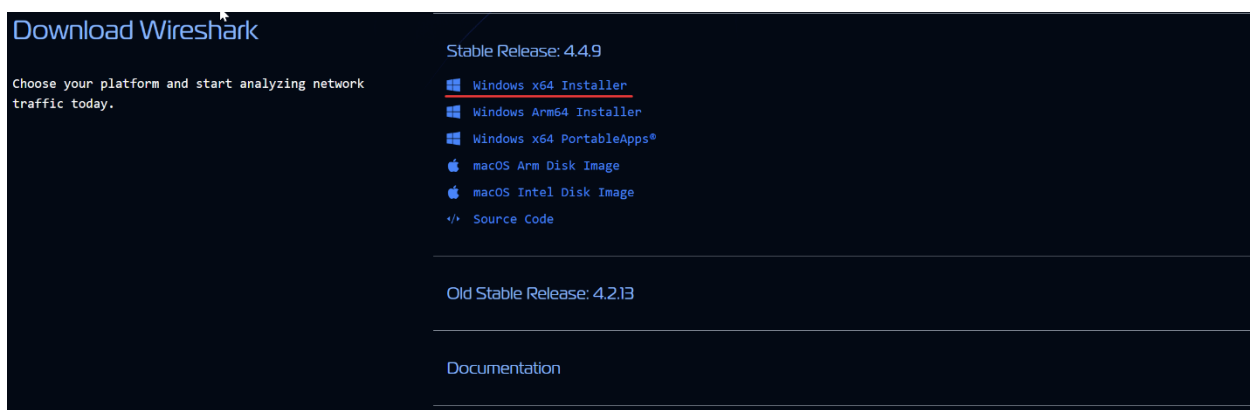
If you need more guidance, please follow the [Cisco Packet Tracer Download and Installation Instructions](#).

**System Requirements:**  
Computer with either Windows (10, 11), MacOS (10.14 or newer) or Ubuntu (20.04, 22.04) LTS operating system, amd64(x86-64) CPU, 4 GB of free RAM, 1.4 GB of free disk space

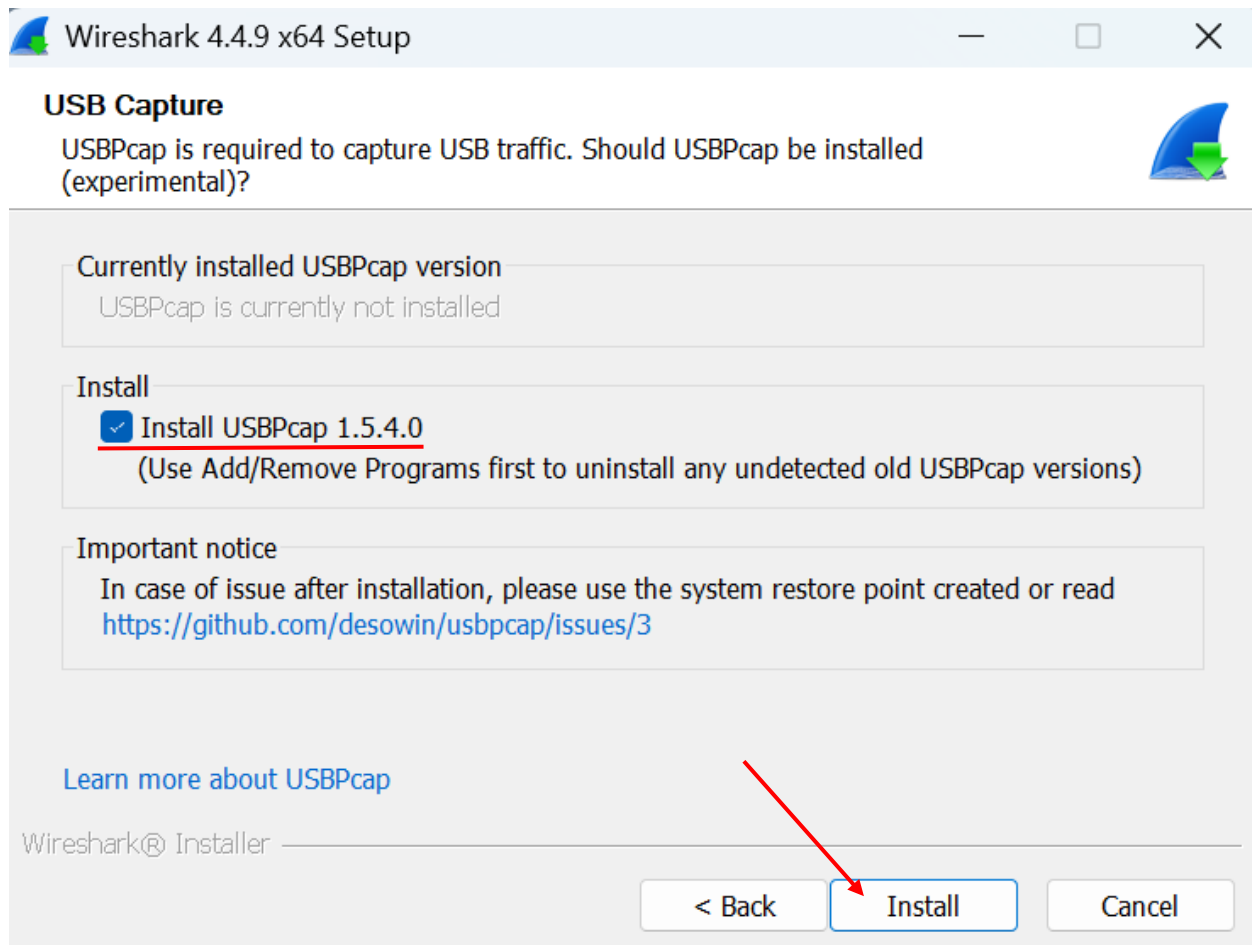
Under Learning Resources, select the Windows 64bit download link. Choose a location and run the setup once the file finishes downloading.



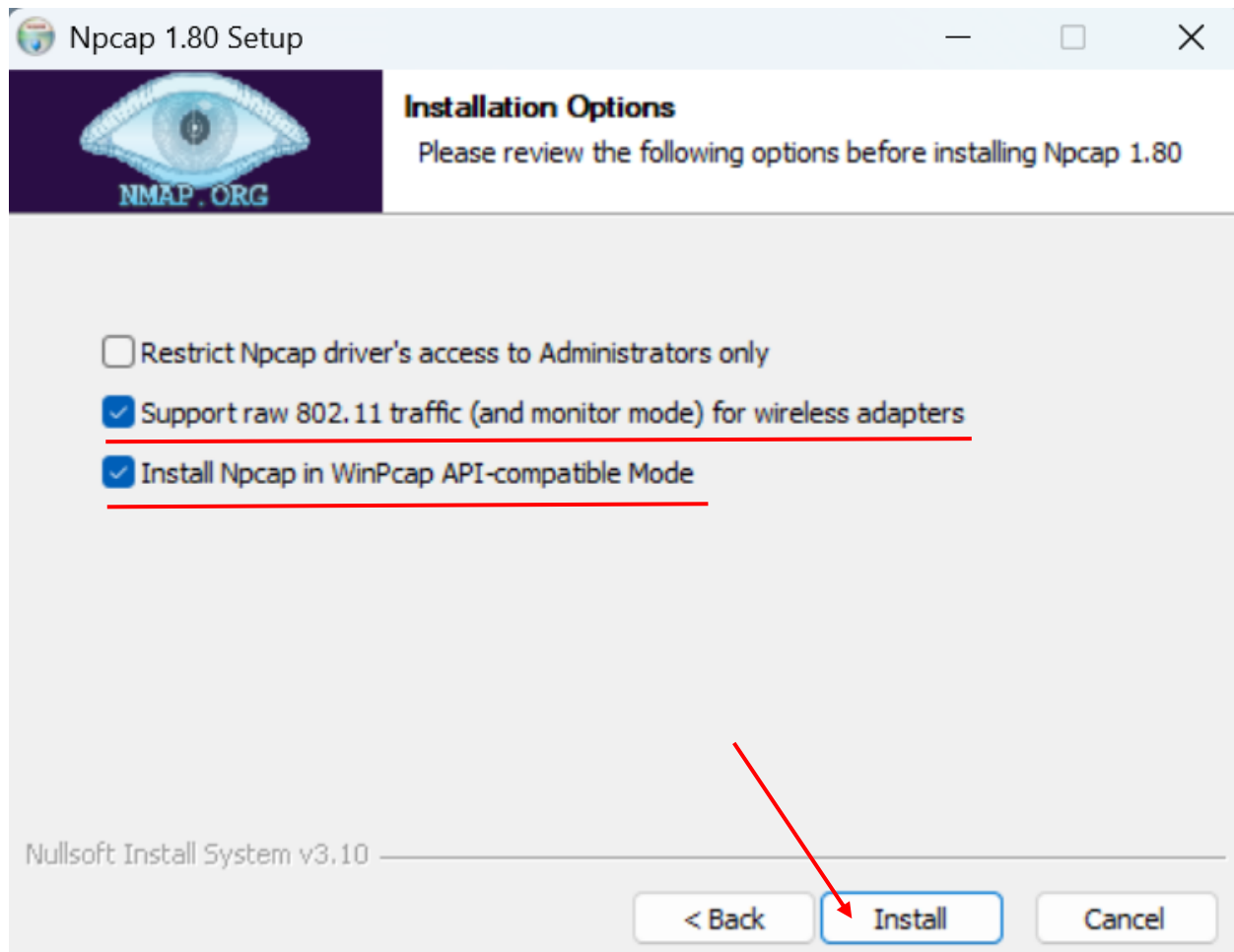
Understand and accept the agreement, then click next. Then, continue to click Next and finally Install.



To download the network protocol analyzer Wireshark, go to <https://www.wireshark.org/#download> or go to the wireshark homepage and click download. Then, select the Windows x64 Installer.



Once the file downloads, open it and ensure Install USBPcap is selected. Then, click Install.



Continue through the installer until you reach installation options regarding Npcap. Select the support for raw 82.11 traffic for wireless adapters and the option to install Npcal in WinPcap API-compatible Mode. After, click Install. Continue through and wait for the install to finish.

## Download

### Latest releases:

| Link                                | Type     | Platform      | Size   | Date       |
|-------------------------------------|----------|---------------|--------|------------|
| <a href="#">rufus-4.9.exe</a>       | Standard | Windows x64   | 2 MB   | 2025.06.15 |
| <a href="#">rufus-4.9p.exe</a>      | Portable | Windows x64   | 2 MB   | 2025.06.15 |
| <a href="#">rufus-4.9_x86.exe</a>   | Standard | Windows x86   | 1.9 MB | 2025.06.15 |
| <a href="#">rufus-4.9_arm64.exe</a> | Standard | Windows ARM64 | 6 MB   | 2025.06.15 |

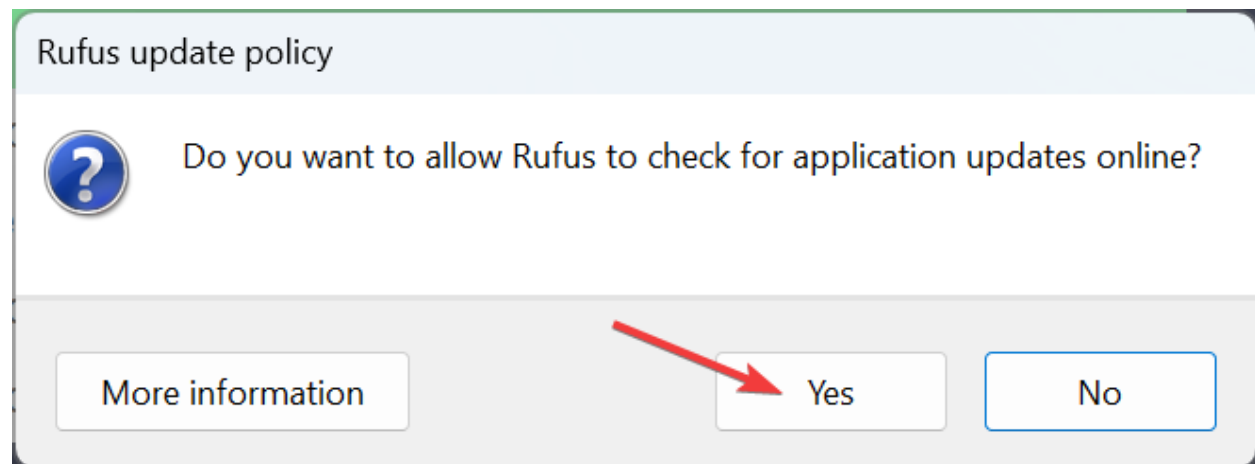
[Other versions \(GitHub\)](#)

[Other versions \(FossHub\)](#)

### System Requirements:

Windows 8 or later. Once downloaded, the application is ready to use.

To install Rufus for making bootable USB flash drives, visit [rufus.ie](https://rufus.ie) and scroll down to the downloads. Click on the latest standard .exe and open the app.



When asked for permission to check for updates online, click Yes.



## Problems

The largest problem I encountered while installing Windows 11 was figuring out what software, programs, and drivers, were essential for the basic setup. This problem was solved by talking with peers. Many people had already gone through this process or knew far more about Windows and drivers than me. By talking with peers, I learned that most drivers are automatically installed, but programs like NVIDIA App can be used to keep drivers up to date. For essential software, I was given the tip to remember the programs used in the first-year Cisco Networking Academy course. This is where I learned of Ninite, making it easy to install the essentials in bulk.

## Conclusion

Installing Windows on a new machine is a straightforward process. Microsoft's directions are clear, making the installation process quick and doable regardless of one's familiarity with technology. Installing essential software depends on the purpose of the machine. For a Cisco networking environment, terminal emulators like PuTTY and protocol analyzers like Wireshark are a necessity. With the operating system set up and the essentials downloaded, this computer is ready for working within a networking space.